Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/30/2007
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2005	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Meck, J@n	**Contact Phone:**	281-244-5405
Contact Email:	janice.v.meck@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: PI moved to University of Washington in summer 2008 and end date changed to 9/30/2008 (from 5/31/2009); new grant will be issued for remainder of this award, per J. Dardano/JSC (8/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Brian (Cleveland Clinic) Licata, Angelo (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNJ05HE72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involve quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on DLS theory. Subjects were randomized to control or exercise groups and confined to 6-degree head-down bedrest in the General Clinical Research Center (GCRC) at the Cleveland Clinic for 12 weeks. During this time, the exercise group underwent individualized daily exercise programs in the unique Zero Gravity Locomotion Simulator (ZLS) at CCF designed to replace their daily mechanical load stimulus experienced during free living. DXA, MRI, and QCT scans were conducted at the start and end of bedrest to assess variations in bone mineral density, changes in muscle volume, and regional changes (cortical vs. trabecular) in bone. Strength measurements were collected pre- and post-bedrest, and urinary and serum markers of bone resorption and formation were also assessed. Dietary intake sufficient to maintain constant body weight with a balanced intake of macro- and micronutrients was planned and supervised by a registered dietician, and a physician was available 24 hr/day to attend to any medical needs. A rehabilitation program was offered to subjects in the 8 weeks following bedrest. The proposed experiment should provide a categorical answer to the question of whether intermittent load replacement can adequately protect the musculoskeletal system against hypokinetic osteopenia and muscle atrophy and may open a new era in individualized exercise countermeasure planning. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in BMD, bone quality, and muscle atrophy during 12 weeks of bedrest. The goals of the proposed study are congruent with the goals identified in the critical path roadmap for countermeasure development. We will address the following Enabling Question 1e of the Bioastronautics Roadmap: What are the specifics of the optimal exercise regimen with regard to mode, duration, intensity and frequency, to be followed during exposure to hypogravity so as to minimize decreases in bone mass? Is impact loading an essential element and, if so, how can it be produced in hypogravity?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The previous two fiscal years have been very productive for this project. During this time GCRC support was obtained and an infrastructure was established to carry out the research. The first four campaigns, (including a total of 4 control and 3 exercise subjects), have been successfully completed. The fifth campaign is currently underway and involves one female subject who is an exercise subject. In collaboration with an NSBRI-funded project, the NASA Glenn Research Center and CCF project team developed a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. The pSLD was used for one exercise subject in campaign 2 and two exercise subjects in campaign 3 and functioned reliably. The exercisers in campaign 3 wore customized harnesses with bilayer-padded shoulder straps. The exerciser in the previous campaign wore a harness with single-layer-padded shoulder straps. The project team has modified existing mathematical theories on daily load stimulus (DLS) incorporating new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. Extensive MATLAB code has been written and is in its final stages of completion, allowing users to analyze complete sets of in-shoe force data and calculate DLS. This revised DLS is being used to formulate the individualized exercise prescriptions used by the exercising subjects. The revised algorithm is currently being implemented into a toolkit for the real-time processing of test subject force-plate data.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh PR, Rice AJ, Davis BL, Carter DR. "A Quantitative Test of On-Orbit Exercise Countermeasures to Bone Demineralization Using a Bedrest Analog." Bioastronautics Investigators' Workshop, Galveston, TX, January 10-12, 2005. Bioastronautics Investigators' Workshop, January 2005. , Jan-2005
Abstracts for Journals and Proceedings	Cavanagh PR, Rice AJ, Englehaupt RK, Kuklis MM, Gopalakrishnan R, Novotny SC, Genc KO, Davis BL, Ilaslan H, Licata AA. "A Quantitative Test of On-Orbit Exercise Countermeasures to Bone Demineralization Using 12 weeks of Bedrest." 26th Annual Cleveland Clinic Research Day, Cleveland, OH, November 30, 2006. 26th Annual Cleveland Clinic Research Day, November 2006. , Nov-2006
Abstracts for Journals and Proceedings	Cavanagh PR, Rice AJ, Novotny SC, Davis BL, Kuklis MM, Genc KO, Gopalakrishnan R, Englehaupt RK. "A Quantitative Test of On-Orbit Exercise Countermeasures to Bone Demineralization Using 12 weeks of Bedrest." NASA Human Research Program Investigators' Workshop, League City, TX, February 12 – 14, 2007. NASA Human Research Program Investigators' Workshop, February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Humphreys BT, Genc KO, Perusek GP, Cavanagh PR. "Algorithm for Identification of Running, Walking and Standing Activity in Foot Force Data." Annual Meeting of the American Society of Biomechanics, Palo Alto, CA, Aug 22 – 25, 2007. American Society of Biomechanics annual meeting, August 2007. , Aug-2007
Abstracts for Journals and Proceedings	Genc KO, Humphreys BT, Perusek GP, Cavanagh PR. "The Enhanced Daily Load Stimulus (eDLS): Accounting for Saturation, Recovery and Standing." Annual Meeting of the American Society of Biomechanics, Palo Alto, CA, Aug 22 – 25, 2007. American Society of Biomechanics annual meeting, August 2007. , Aug-2007